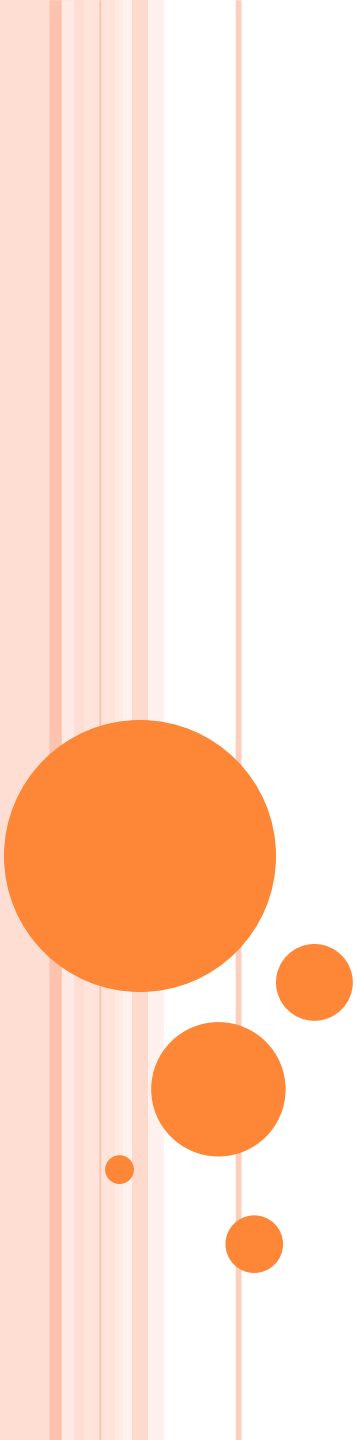




PROJECT TITLE : SERVERLESS CLOUD DICTIONARY APPLICATION

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AGENDA

- Problem Statement
- Solution Overview
- Key Features
- AWS Services Used
- Architecture
- Working Process
- Advantages
- Future Work
- Conclusion



PROBLEM STATEMENT

- Cloud computing includes many technical terms that beginners find difficult to understand.
- Searching for definitions across multiple websites is time-consuming.
- There is no simple, centralized application dedicated to quickly looking up cloud terms.
- Users need an easy and fast way to learn cloud computing concepts.



SOLUTION OVERVIEW

- The Serverless Cloud Dictionary Application provides a simple web interface where users can search for cloud computing terms such as **AWS Lambda, AWS CloudFormation, and Virtualization.**
- The application uses AWS serverless services to retrieve definitions instantly without requiring server management.



KEY FEATURES

- Search cloud computing terms
- Instant definition retrieval
- Fast and user-friendly interface
- Serverless architecture
- Scalable and reliable
- Secure cloud-based storage
- Easy to maintain



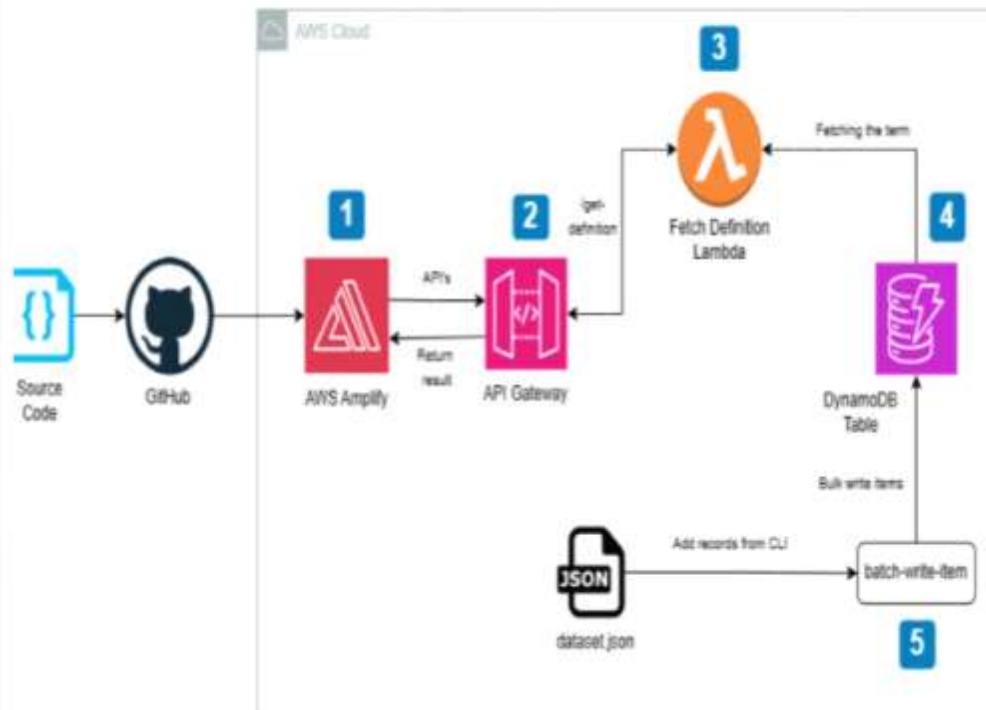
AWS SERVICES USED

AWS Service	Icon	Purpose
AWS Amplify		• Hosts the React web application.
Amazon API Gateway		• Receives user requests and routes them to AWS Lambda.
AWS Lambda		• Processes search requests and retrieves definitions.
Amazon DynamoDB		• Stores cloud terms and their definitions.
AWS IAM		• Manages user access and permissions securely.



ARCHITECTURE

SERVERLESS CLOUD DICTIONARY APPLICATION



1 The frontend React application is hosted on AWS Amplify, allowing users to search for terms and view their definitions through a web interface.

2 The frontend sends HTTP requests (GET) to API Gateway, which exposes an endpoint (/get-definition) to manage interactions with the backend.

3 The API Gateway triggers the **Fetch Definition Lambda** function for GET requests, which fetches the corresponding term's definition from the DynamoDB table.

4 DynamoDB stores cloud terms and their definitions. It serves as the database for retrieving and storing term data via Lambda functions triggered by API requests.

5 The batch upload process (via CLI or script) utilizes batch-write-item to insert multiple terms and definitions into the DynamoDB table in bulk.

WORKING PROCESS

- User enters a cloud term.
- React application sends the request.
- API Gateway receives the request.
- AWS Lambda processes it.
- Lambda searches DynamoDB.
- The definition is retrieved.
- The result is displayed to the use



ADVANTAGES

- No server management
- Automatic scaling
- Low operational cost
- Fast response time
- High availability
- Easy maintenance
- Supports cloud learning



FUTURE WORK

- Voice-based search
- Multi-language support
- User authentication
- Favorite terms
- Search history
- AI-powered explanations
- Word of the Day
- Quiz for cloud concepts



CONCLUSION

The **Serverless Cloud Dictionary Application** is an efficient learning tool that helps users quickly understand cloud computing terminology. By using AWS serverless services, it provides a scalable, cost-effective, and reliable solution without requiring server management.



*Thank
you!*

